111 Essential Proofs Every CMI Aspirant Should Know

Mohit Dua

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1 Introduction

These proofs will (hopefully) help aspirants collect various strategies to solve Olympiad/CMI-level problems. Please note that, I've completely omitted Euclidean Geometry as part of my preparation for it is too complex for my tiny brain to handle. Note that indices marked * are not essential (but check it out if there's enough time).

2 Combinatorics

1. Pascal's Rule
2. Number of different pairings of a $2n$-element set A. (Example 1.4.4 CCK)
3. Recursive formula for Stirling Numbers of First Kind
4. Show that if $|X| = n$, then, $|P(X)| = 2^n$.

* Show that $(4n)!$ is a multiple of $2^{3n}3^n$ (Example 1.6.6 CCK)

5. Number of r-element multi-subsets of M, where $M = \{\infty \cdot a_1, \infty \cdot a_2, \ldots, \infty \cdot a_n\}$ - Stars And Bars Method
6. Recursive formula for Stirling Numbers of Second Kind
7. Basic Identities involving $\binom{n}{k}$
8. Combinatorial Identities (including Vandermonde's Identity, Hockey-stick Identity)
9. Reflection Principle
10. Proof of IEP (set form and properties form)
11. Number of Surjective Functions - Formula for $S(n, m)$ [2nd kind]
12. Number of Derangements $D(n, r, k)$

* Solution of a General Linear Homogeneous Recurrence Relation

3 Number Theory

1. Division Algorithm (existence and uniqueness of q, r with constraints)
2. Standard Divisibility Results
3. Bezout's Lemma
4. $\gcd(a, b) = 1 \iff \exists x, y \in \mathbb{Z}$ s.t. $ax + by = 1$ (and its corollaries)
5. Euclid's lemma (if $a \mid bc$ and $\gcd(a, b) = 1$, then, $a \mid c$)
6. Euclidean Algorithm (finding gcd using division algorithm)
7. $\gcd(a, b) \operatorname{lcm}(a, b) = ab$ for positive integers a, b.
8. Condition for solvability of a diophantine equation - proof for existence of the proposed set of solutions.
9. Fundamental Theorem of Arithmetic.
10. Proof of irrationality of $\sqrt{n}$ where n is not a perfect square (for example $\sqrt{2}$)
11. Proof for infinite number of primes - Euclid
12. $a \equiv b \pmod{n} \iff a$ and b leave the same nonnegative remainder when divided by n
13. Standard Congruence Results
14. If $ca \equiv cb \pmod{n}$, then, $a \equiv b \pmod{n/d}$ where $d = \gcd(c, n)$.
15. Condition for solvability of a linear congruence
16. CRT
17. Condition for unique solution of a system of linear congruences.
18. Fermat's Theorem (and corollary)
19. Wilson's Theorem

   * The quadratic congruence $x^2 + 1 \equiv 0 \pmod{p}$, where p is an odd prime, has a solution $\iff p \equiv 1 \pmod{4}$

20. Formula for: $\tau(n)$ and $\sigma(n)$ where $n = p_1^{d_1} p_2^{d_2} \dots p_n^{d_n}$ and $n > 1$.
21. Legendre's Formula
22. Euler's Function properties ($\varphi(p^k)$ and multiplicative nature)
23. Euler's Theorem.

4 Analysis

Limits

1. Uniqueness of Limits
2. Every convergent sequence is bounded
3. Algebraic Limit Theorem
4. Order Limit Theorem
5. Squeeze Theorem (for sequences)

* Cesaro Means

6. MCT
7. Cauchy Condensation Test
8. p-series Test
9. $\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n$ (Separate strict inequality and equality cases)
10. For $a_n > -1$, $\prod_{n=1}^{\infty}(1+a_n)$ converges $\iff \sum_{n=1}^{\infty} a_n$ converges
11. Subsequences of a convergent sequence converge to the same limit as the original sequence

* Bolzano-Weierstrass Theorem

12. Cauchy Criterion
13. ALT for series
14. Cauchy Criterion for series
15. Comparison Test
16. Absolute Convergence Test
17. Alternating Series Test
18. Ratio Test
19. Root Test
20. If a series converges absolutely, then any rearrangement of this series converges to the same limit
21. Sequential Criterion For Functional Limits
22. Algebraic Limit Theorem For Functional Limits
23. $\lim_{x\to a} f(x) = L$ iff both the right and left limits equal L
24. Squeeze Theorem (for functions)

Continuity

25. Algebraic Continuity Theorem
26. Composition Of Continuous Functions
27. Contraction Mapping Theorem
28. Intermediate Value Theorem (IVT) - using AoC
29. Continuous Function $\implies$ IVP; but, IVP $\not\Longrightarrow$ Continuous Function (Counterexample - with proof)
30. Monotone + IVP $\implies$ Continuous Function

Differentiation

31. Differentiability $\implies$ Continuity
32. Algebraic Differentiability Theorem
33. Chain Rule
34. Darboux's Theorem
35. Rolle's Theorem
36. Mean Value Theorem
37. Generalized Mean Value Theorem
38. L'Hospital's Rules
39. Taylor Series

Integration

40. Fundamental Theorem of Calculus (I, II)
41. Integration By Parts
42. Linearity of the Integral
43. Mean Value Theorem for Integrals
44. Comparison Theorem for Integrals

5 Inequalities

Also consider the cases when equality is obtained.

1. AM–GM Inequality
2. Cauchy-Schwarz Inequality
3. Rearrangement Inequality

4. Chebyshev Inequality
5. Jensen's Inequality
6. Schur's Inequality
7. Elementary Symmetric Polynomial Identities

6 Polynomials

1. Factor Theorem
2. Remainder Theorem
3. Vieta's Formulae
4. Polynomial Identity Principle
5. A polynomial of degree n has at most n roots
6. Uniqueness of polynomial interpolation through $n + 1$ points
7. Lagrange Interpolation Formula
8. Newton's Identities (relations between power sums and symmetric sums)
9. Rational Root Theorem
10. Descartes' Rule of Signs

7 Complex Numbers

1. Basic identities involving complex conjugates ($z + w$, zw, $|z|^2 = z\bar{z}$)
2. Triangle Inequality $|z + w| \leq |z| + |w|$
3. Reverse Triangle Inequality $||z| - |w|| \leq |z - w|$
4. Polar representation of complex numbers ($z = r(\cos\theta + i \sin\theta)$)
5. De Moivre's Theorem
6. The nth roots of unity form a regular n-gon on the unit circle
7. Sum of the nth roots of unity is 0
8. Factorization of $x^n - 1$ using roots of unity
9. Every complex polynomial of degree n has exactly n complex roots (Fundamental Theorem of Algebra)
10. Geometric interpretation of complex multiplication (multiplication corresponds to rotation and scaling)

8 Fundamental Proof Techniques

The following techniques appear repeatedly in Olympiad and CMI-level problems. Recognizing when to apply these methods is often the key to solving difficult problems.

1. **Proof by Contradiction** Assume the negation of the desired result and derive a logical contradiction.
2. **Mathematical Induction** Prove a base case and then show that if the statement holds for n, it holds for $n + 1$.
3. **Strong Induction** Assume the statement holds for all values up to n to prove it for $n + 1$.
4. **Infinite Descent** Assume a smallest counterexample exists and derive a smaller one, producing a contradiction.
5. **Construction Method** Construct an object with desired properties to prove existence (e.g., Euclid's proof of infinitely many primes).
6. **Extremal Principle** Consider the smallest or largest element in a set satisfying certain conditions.
7. **Pigeonhole Principle** If more objects than containers exist, at least one container must contain multiple objects.
8. **Double Counting** Count the same set in two different ways to obtain an identity.
9. **Invariants** Identify a quantity that remains unchanged under certain operations.
10. **Monovariants** Identify a quantity that consistently increases or decreases under operations.
11. **Symmetry Arguments** Exploit symmetry in variables or structures to simplify reasoning.
12. **Parity Arguments** Use properties of even and odd numbers to restrict possible cases.
13. **Bounding** Establish upper or lower bounds to constrain possible values.
14. **Case Analysis** Divide the problem into a finite number of cases and analyze each one.
15. **Substitution** Introduce new variables or expressions to simplify equations or functional relations.
16. **Working Modulo** n Use modular arithmetic to simplify divisibility or congruence relations.
17. **Factorization** Rewrite expressions to reveal hidden structure or divisibility.
18. **Algebraic Manipulation** Transform expressions through identities or rearrangements to reveal structure.
19. **Geometric Interpretation** Interpret algebraic relations geometrically (especially useful in complex numbers).
20. **Recursive Thinking** Relate a problem of size n to smaller instances of the same problem.